

Melbourne Airbnb Market Screening

Interim Project Report

CMCE30005 Business Analytics Challenge | Semester 2 2026 | TheNextChapter Group 2

Eric Huang, Loc Le, Qihang Sun and Maksym Xu

Repository: <https://github.com/Qihang-cpu/CMCE30005-TheNextChapter>

Introduction

Our client leases residential properties and runs them as Airbnb accommodation (“rental arbitrage”), which needs lease and council permission. With limited start-up funds, the client must choose which Melbourne areas and property types to investigate before signing leases and buying furniture; a poor choice ties up money and delays the business.

We use the Inside Airbnb Melbourne snapshot of June 2026 supplied through the subject’s LMS (Inside Airbnb, 2026): property details, locations and dated guest reviews. Because a review can only follow a completed stay, dated reviews are a countable record of past guest activity; the calendar cannot separate bookings from host-blocked dates.

Our output is a shortlist of segments for further investigation, based on past review activity, plus a test of whether a model improves on segment averages.

Problem Definition and Objectives

Research question: Which types of Melbourne property — defined by council area (LGA), apartment or house, and number of bedrooms — most often reach a high level of guest-review activity, and does a statistical model add anything to simple segment averages when ranking them? “High” means at least 30 reviews in the year before the data were collected, the top quarter of a separate reference group of comparable listings.

Hypotheses: (H1) segments differ materially in the share of listings reaching that level; (H2) a model using listing attributes ranks listings better than the segment average alone.

Objectives: (1) a segment comparison table; (2) a model-versus-average test; (3) a shortlist with uncertainty ranges.

We study entire homes with one to three bedrooms, what a small operator can furnish and let whole; rental units and condos form the apartment group, homes and townhouses the house group. Each segment needs at least 50 listings as minimum comparison support (host counts and resampled ranges are reported); quoted prices of AUD 30–1,500 are an exploratory boundary; and a first review at least 365 days before collection gives at least one year of observable review history, not proof of continuous operation.

We call the share of listings reaching 30 reviews “attainment”; 30 reviews means at least 30 reviewed stays, roughly one every 12 days. Attainment measures review activity, not bookings, occupancy or profit.

Data Description

The dataset contains 25,728 listings, 9,390,720 calendar records and 1,026,690 reviews, collected between 17 June and 1 July 2026. We use LGA, property type, bedrooms, capacity, bathrooms, amenities and review dates. Its modelled occupancy and revenue fields are built from price, minimum stay and review counts, so ranking on them would only reproduce that construction; we rank on review counts directly.

Cleaning converts price text to numbers, standardises missing values and extracts bathroom and amenity counts; price is missing for 6,553 listings and bedrooms for 4,679. The supplied last-12-months review field spans 366 dates, so we rebuilt counts over exactly 365 days from review dates (Table 1).

Table 1. Sample funnel

Stage	Listings
All listings in the dataset	25,728
Entire homes, 1–3 bedrooms	16,576

Stage	Listings
Standard dwelling types	14,895
Quoted price AUD 30–1,500	11,099
First review ≥ 365 days before collection	6,891
Analysis sample (50+ per segment, reference hosts excluded)	3,873
Reference group (separate, from the 6,891)	906

The reference group's 75th percentile is 29.75 reviews, giving a threshold of 30. The analysis sample (3,873 listings from 1,726 hosts) has 14 segments across six LGAs, about 64% in Melbourne LGA, and keeps 334 listings with zero annual reviews. Overall, 981 listings (25.33%) reach the threshold, and attainment ranges from 8.25% to 36.60% across segments (Figure 1).

Review activity across established residential segments

Common upper-quartile target: at least 30 reviews | 3,873 analysis listings in 14 segments

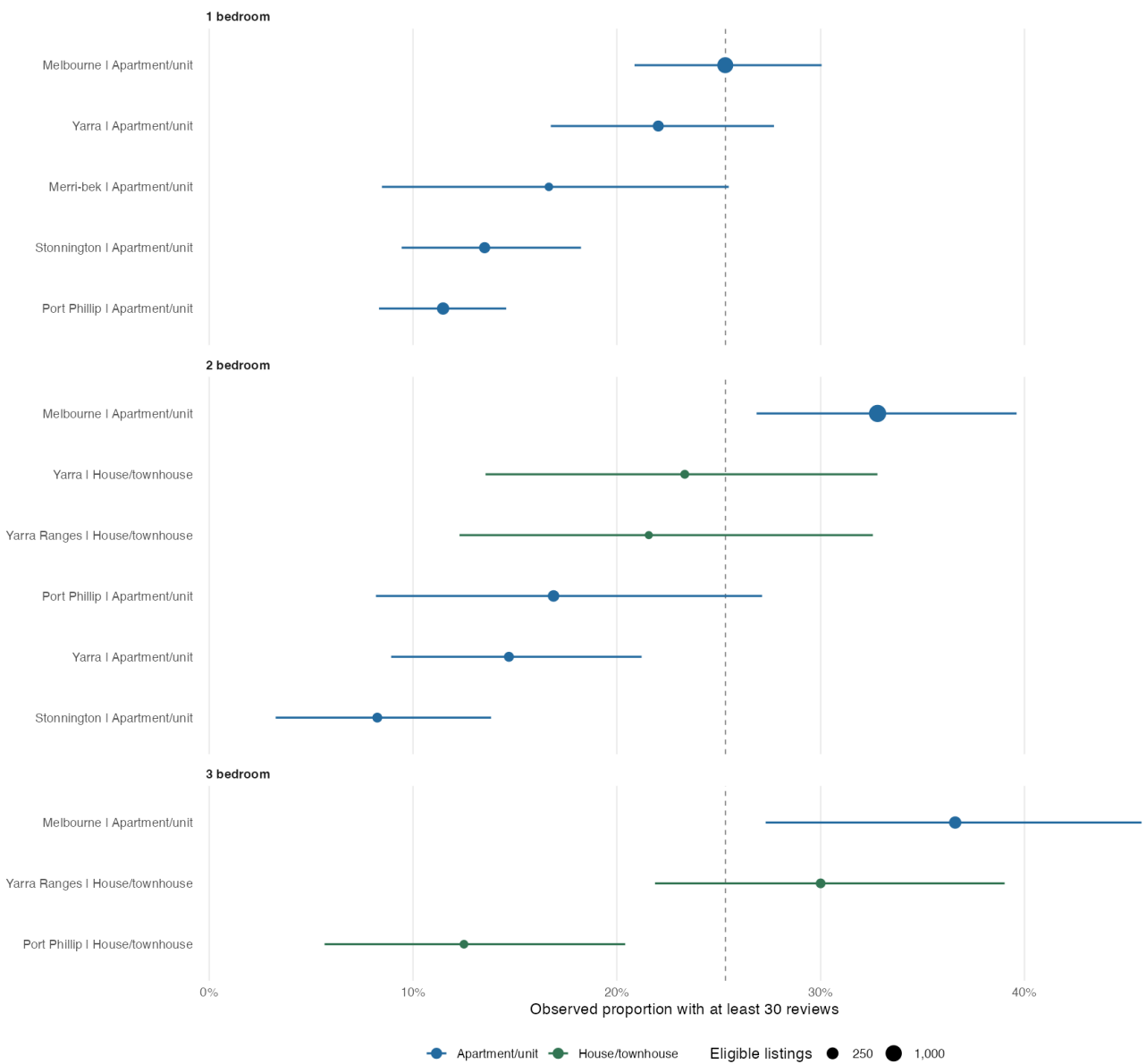


Figure 1. Share of listings reaching 30 reviews by segment; lines are 90% host-resampled ranges, dashed line the pooled 25.33% rate.

Of the 6,801 listings with no usable price (missing or outside AUD 30–1,500), 82% had no review in the past year, so the priced sample is the more active part of the market. Listings that left Airbnb before collection are absent, which limits generalisation to new entrants.

Methodology and Analytical Approach

A fixed rule on host identifiers sets aside about 20% of hosts as the reference group that defines the 30-review target; they are excluded from all training and testing. We use five-fold validation, each part tested in turn on a model trained on the other four, with each host's properties kept in one fold so a host with many near-identical apartments cannot sit in both training and test data (scikit-learn developers, n.d.).

Logistic regression is the simple reference; random forest and gradient boosting capture nonlinear patterns. Basic features are location, configuration, capacity, bathrooms and amenity count; an enhanced set adds coordinates, beds and nine amenities. We compare six variants of these models, with and without probability calibration and with and without current price and minimum stay. Review-based predictors, ratings and host badges are excluded.

Every model is compared with a segment-average baseline: each listing gets its segment's attainment among training hosts. AUC is how well the model orders listings that reach the target above those that do not; 0.5 is a coin toss. Average precision is how many of its top picks really reach the target. Brier score is how far its percentages are from what happened; lower is better. R handles cleaning and tables; Python (scikit-learn 1.7.2) runs the models.

No untouched hold-out set exists (model choice used the same folds; item (1) below re-tests it), and property details may have changed during the outcome year.

Analysis Plan and Progress to Date

Following Week 5 feedback, we dropped the earlier rent-versus-revenue (ROI) design, which needed external rental data, and now compare only within the snapshot. Cleaning, review reconstruction, descriptive analysis and validation are complete (Table 2, same folds).

Table 2. Predictive performance

Predictor	AUC	Average precision	Brier score
Segment average (training hosts)	0.581	0.285	0.1868
Basic property- only logistic	0.573	0.285	0.1886
Enhanced property-only random forest	0.640	0.350	0.1813

The enhanced random forest raises AUC by 0.059 and lowers Brier by 0.0055, beating the segment average in all five folds; across 20 further host splits it wins on AUC every time (mean 0.626, range 0.589–0.648) and on Brier in 16. Adding current price and minimum stay lifts boosting to AUC 0.749 and Brier 0.162, but these are present-day settings, not pre-opening attributes, so they stay out of the shortlist.

Table 3. Initial search candidates (forest predictions from models trained without each listing's host)

Segment	Listings / hosts	Observed attainment	Predicted probability	95% range
Melbourne 3BR apartments	306 / 139	36.60%	34.45%	33.0–35.8%
Melbourne 2BR apartments	1,235 / 482	32.79%	31.41%	29.6–33.0%
Yarra Ranges 3BR houses	90 / 77	30.00%	27.82%	26.1–29.3%

The ranges resample hosts with predictions held fixed, exclude model refitting, selection and threshold uncertainty, and are much narrower than Figure 1's observed-rate ranges. H1 holds at the extremes (8.25% against 36.60%) but neighbouring segments overlap; H2 holds on ranking metrics, not on the shortlist, since the model changes no candidate — the segment comparison carries the decision and the model adds a supplementary historical score. The same three candidates lead in 19 of 20 host splits, under support rules of 30, 50 and 75 listings, and without the history or price rule; only their order changes.

An out-of-time check on 2,657 listings (1,208 hosts, 10 segments, two years of reviews) uses two non-overlapping 365-day windows and a 34-review target set from the reference hosts' earlier window. The earlier year's top three segments reached it in 27.1% of listings the following year against 14.9% elsewhere, a 12.3-point lead (host-resampled 95% range 5.1–20.1), though the exact top-three set recurred in only 51.5% of resamples: a persisting historical advantage for the candidate set, not a new operator's profit.

The client can use the segment averages as a first screen, prioritise lease enquiries there, and verify permissions, rent, costs and availability before signing.

Remaining work:

1. Select the model within training data and re-test it — Eric Huang (modelling lead), Week 9 (25 Sep).
2. Re-rank with uncertainty ranges under wider property-type rules — Maksym Xu (data lead), Week 10 (2 Oct).
3. One-page client brief with search priorities and pre-lease checklist — Loc Le (report lead), Week 11 (by the 13 Oct presentation).
4. Final report and presentation — all, Qihang Sun coordinating, Weeks 11–12.

References

Inside Airbnb. (2026). Melbourne listings, calendar and reviews [June 2026 dataset supplied through CMCE30005 LMS].

Scikit-learn developers. (n.d.). Cross-validation and probability calibration.
https://scikit-learn.org/stable/modules/cross_validation.html

OpenAI. (2026). ChatGPT/Codex assistance with project planning, report drafting and code review [Generative AI outputs, September 2026].

Anthropic. (2026). Claude Code assistance with analysis-code development [Generative AI outputs, September 2026].

AI Use Acknowledgement

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